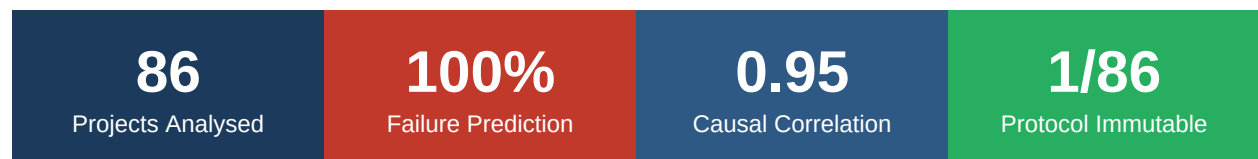


BLOCKCHAIN RISK EVALUATION MODEL

Structure × Process × Persistence

A Coordination System Risk Model



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BREM — Blockchain Risk Evaluation Model

March 2026

Confidential — For Evaluation Purposes

Contents

Executive Summary

Over \$60 billion in value has been destroyed across blockchain initiatives — from enterprise consortia that quietly shut down to DeFi protocols that collapsed overnight. Most of it was preventable.

Across 86 blockchain implementations analysed — spanning enterprise, institutional, and DeFi deployments — **100% of failures** above the 2.5 risk threshold follow three causal cascades. These cascades can be detected before a single line of code is written.

The Blockchain Risk Evaluation Model (BREM) provides the analytical framework for identifying these cascades. It evaluates blockchain systems through the lens of **Structure, Process, and Persistence** across three coordination domains: machines interacting with machines (Network), humans interacting with machines (System State), and humans interacting with humans (Law).

The core finding is architectural determinism. Successful systems minimise off-ledger coordination at critical points. Failures occur when correctness, scaling, or enforcement depends on external negotiation, special permissions, or institutional intervention. The industry treats legal classification as a regulatory question. In reality, it is an architectural question.

BREM's scoring distinguishes between architectures that can scale to global adoption and those that cannot. This is not a preference. It is an engineering constraint. Architectures built on shared mutable global state face serialisation pressure that grows with write conflicts and hotspot contention, forcing effective sequentiality under load. Architectures built on independent record-level state allow parallelism by construction. These are not comparable approaches to the same problem. One admits pipelined, horizontally scalable transaction processing. The other requires fragmentation (Layer 2s, sharding, rollups) that breaks atomic composability and the chain of title. The model measures this structural ceiling, not a philosophical preference for one community over another.

The empirical evidence is strongly directional and consistent with independent case evidence. The correlation between Architecture and Scalability scores is $r=0.953$. Between Scalability and Economic Fitness, $r=0.916$. Production systems average 1.85 across all cells. Failed projects average 3.84. The gap between production systems and failures reveals a threshold — projects scoring above approximately 2.5–3.0 show sharply elevated failure rates, with no counter-examples of poor architecture achieving economic sustainability.

Three failure cascades account for every project failure in the dataset: the Architecture Death Spiral (architecture constrains scalability, which constrains economic viability), the Mutation Authority Collapse (unbounded mutation authority undermines execution and legal coherence), and the Regulatory Kill (the system fails at the legal coordination layer regardless of technical merit). Only one project — Diem/Libra — had adequate architecture and still failed, killed by sovereign regulatory threat.

The Problem: \$60 Billion in Preventable Failures

Blockchain has a pattern recognition problem. Enterprise consortia burn through hundreds of millions before discovering architectural dead ends. DeFi protocols destroy tens of billions in weeks through the same mutation authority failures wearing different costumes. Projects fail repeatedly for the same structural reasons, yet the industry lacks a systematic framework for identifying these patterns before capital is committed.

Metric	Finding
Projects analysed	86 across enterprise, institutional, and DeFi
Total tracked value destroyed	\$60+ billion
Failure rate (confirmed failures)	38 of 86 (44%)
Predictive accuracy at ≥ 2.5 threshold	100% (zero false negatives)
Architecture as entry point	75% of enterprise failures; reversed in DeFi (sf $\rightarrow$ sm)
Predictive correlation (na $\rightarrow$ ns)	$r = 0.953$
Score gap: Survivors vs Failures	Clean separation, zero overlap across 86 projects

The gap between successful and failed projects is not gradual — it is a cliff. There is no cluster of projects that are “slightly risky but might work.” Either the architecture constrains the system sufficiently, or it does not.

The BREM Model: SPP × SPP

Structure × Process × Persistence

Complex coordination systems must resolve three problems simultaneously: structure (how the system is built), process (how actors interact with it), and persistence (whether outcomes endure over time). Blockchain systems encounter these problems at three distinct coordination layers.

BREM evaluates each layer through the same SPP lens, producing a 3×3 evaluation matrix.

	Structure	Process	Persistence
Network (M2M)	Architecture (na)	Consensus (nc)	Scalability (ns)
System State (H2M)	Execution (se)	Mutation Authority (sm)	Economic Fitness (sf)
Law (H2H)	Standing (ls)	Remedy (lr)	Liability (lp)

Why SPP × SPP

The matrix is SPP applied twice — once vertically to define the rows, and once horizontally to interrogate each row. The vertical application comes first and is the more fundamental of the two.

The rows are SPP. Network (M2M) is the **structure** of the coordination system — the protocol architecture, the consensus mechanism, and the throughput ceiling that determine what the system physically is. System State (H2M) is the **process** operating on that structure — the execution environment, the mutation authority governing who can change the rules, and the economic fitness that determines whether the process is sustainable. Law (H2H) is the **persistence** question — can this system survive contact with the real world of legal standing, judicial remedy, and liability allocation? A system that scores well on Network and System State but fails on Law will be destroyed by regulation, litigation, or unenforceable obligations. The rows answer: what is the system, how does it operate, and will it be allowed to exist?

The columns are also SPP. Within each row, the same lens is applied again. **Structure** is what exists at that layer: the network architecture (na), the execution environment (se), or legal standing (ls). **Process** is how interactions occur at that layer: consensus mechanism (nc), mutation authority (sm), or legal remedy (lr). **Persistence** is the survival question at that layer: can it scale (ns), is it economically viable (sf), or will liability destroy it (lp)?

This double application is not decorative. It captures a structural reality: the same three questions that determine whether a coordination system is viable at the macro level (what is it, how does it work, will it persist) are the same three questions that determine risk at each individual layer. Persistence is always the fatal column — it asks the question that kills. At the Network layer, persistence is scalability: if the architecture hits a throughput ceiling, the system dies regardless of the business plan. At the System State layer, persistence is economic fitness: if mutation authority degrades the value proposition, the system starves. At the Law layer,

persistence is liability: will a regulatory kill shot, uninsurable risk, or unenforceable obligation end the project? All three are fatal. The model measures proximity to each.

Each row also maps to a coherent discipline: Network maps to computer science and distributed systems, System State maps to institutional economics and governance design, and Law maps to private law and equity. This means that each row can be evaluated by domain specialists working within their own expertise, while the model integrates their assessments into a unified risk profile.

The 0–4 Scale: Domain-Specific Scoring

BREM uses a single 0–4 numeric scale, but the dimension being measured is calibrated to each coordination domain. Network measures protocol determinism — how fully the protocol constrains outcomes. System State measures mutation authority — the scope of discretion that has been introduced into what should be a fixed set of rules and data structures. Law measures enforceability — whether outcomes survive contact with the legal system.

Network (M2M): Protocol Determinism

The Network domain measures how fully the protocol constrains outcomes without external layers, special committees, or operational intervention. A score of 0 means the machine guarantees correctness by construction. A score of 4 means outcomes depend on mutable shared state, sequential bottlenecks, and frequent exceptional handling.

Score	Architecture (na)
0	UTXO-style independent spends; contention resolved at record level; no shared mutable state; verifiable without global state
1	Mostly deterministic; constraints are formal and stable; bounded operational choices within protocol rules
2	Deterministic core but relies on middleware, coordinators, or recommended patterns for normal operation
3	Scaling or security depends on multi-party coordination or adjunct layers (L2/bridges required, not optional)
4	Shared mutable global state; sequential execution bottleneck; contention is systemic; outcomes depend on implementation choices

Score	Consensus (nc)
0	Objective work-based ordering (PoW) without artificial throughput constraints; finality determined by energy expenditure; competitive incentive to invest in infrastructure (compute, bandwidth, storage) creates endogenous scaling pressure
1	Formal protocol rules with narrow parameterisation; consensus participants have limited operational discretion; infrastructure investment incentives mostly intact
2	Hybrid mechanisms; some governance coupling to validator selection but bounded by protocol rules; mixed incentive structure
3	Stake-weighted voting with delegation; returns proportional to weight not compute; governance decisions affect security (slashing, validator sets); concentration risk via liquid staking
4	Subjective/social finality; heavy governance coupling to consensus security; committee-based block production; no competitive pressure to scale infrastructure; artificial throughput constraints justified by nebulous decentralisation claims

Consensus mechanism determines scalability through incentive structure, not merely through technical parameters. In proof-of-work without artificial block constraints, miners who invest in infrastructure (compute, bandwidth, storage) process more transactions and earn more

revenue. This creates endogenous scaling pressure: the competitive dynamic drives continuous infrastructure investment. The claim that PoW cannot incentivise throughput scaling assumes artificially constrained block parameters — when those constraints are removed, the incentive to scale is inherent in the reward mechanism.

Proof-of-stake inverts this incentive: returns are proportional to stake weight, not infrastructure investment. The rational actor hoards stake rather than builds capacity. Capacity expansion under PoS depends on protocol upgrades and coordination rather than competitive pressure. Known concentration vectors — liquid staking pools, validator cartels, and governance token accumulation — compound this dynamic.

The industry commonly justifies artificial throughput constraints by invoking “decentralisation,” but this term is rarely defined operationally. Paul Baran’s foundational network theory defines decentralisation as path redundancy and survivability — resistance to single points of failure through multi-link connectivity, not raw node count. A network achieves survivability with sufficient redundancy in links and routing paths, not by maximising the number of participants running identical software. Regulatory definitions reinforce this: the SEC’s safe harbour framework defines decentralisation as the absence of economic or operational control by any single entity, not as a node-count metric. Treating “more nodes equals more decentralised” as an axiom conflates an implementation detail with a network property, and provides rhetorical cover for architectures that cannot scale.

Score	Scalability (ns)
0	Scales via native parallelism without mandatory off-chain fragmentation; pipeline architecture with measurable stage capacities
1	Linear scaling with hardware; minor coordination overhead; no mandatory fragmentation
2	Scales within bounds but requires operational tuning; some throughput ceiling from protocol design
3	Requires fragmentation/bridges/L2 as baseline operation; throughput limited by base-layer protocol
4	Hard protocol-level throughput ceiling; scaling requires fundamental architectural changes or mandatory off-chain layers

System State (H2M): Mutation Authority

The System State domain measures what has authority to cause state transitions — both data-level inputs and protocol-level changes. A blockchain is a complete economic system: the protocol, the node operators, and the users. The critical question is the scope of mutation authority: who can write data to the system, who can change the execution environment, who can modify network parameters, and who can mutate the protocol itself. A protocol should be a fixed set of rules and data structures. Mutation authority measures how much discretion has been introduced into that fixed system.

Score	Execution (se)
0	Bounded execution model; no halting problem; no reentrancy class; no bridge/protocol dependencies; deterministic script execution
1	Safe execution with formal constraints; upgrade paths are protocol-governed; minimal attack surface
2	Turing-complete execution with safeguards; some reentrancy risk mitigated by patterns; bridge dependencies for specific use cases
3	Execution environment suffers known vulnerability classes; requires constant monitoring; bridge protocols as operational dependency
4	Halting-problem exposure; reentrancy attacks demonstrated; state explosion risk; bridge failures cause systemic loss; upgrade hazards

Score	Mutation Authority (sm)
0	Protocol immutable; no party can mutate core rules or data structures; data ingress governed by deterministic validation; operational governance via legal framework without protocol mutation; oracle inputs validated by protocol rules
1	Mutation scope narrow and formally constrained; only off-chain parameters adjustable; data input rules fixed by protocol; decision authority clearly allocated with minimal discretion
2	Network parameters mutable within protocol-defined bounds (gas limits, block size); oracle/data feed rules require some coordination; moderate governance overhead for operational decisions
3	Execution environment and economic rules mutable (VM upgrades, fee structures); protocol changes achievable through governance process; data input rules subject to committee decisions; authority unclear; cartel capture risk
4	Core protocol arbitrarily mutable; any rule or data structure can change through governance; consensus rules, architecture, and scalability subject to discretionary mutation (soft forks, hard forks); data ingress rules politically determined; upgrade authority concentrated

Score	Economic Fitness (sf)
0	Unit economics viable at scale without subsidy; fee predictability; micropayment-viable; cost-to-verify proportional to value
1	Economically coherent; sustainable fee model; minor scaling friction; viable for most business

	models
2	Economics work for high-value transactions; micropayments not viable; some subsidy or cross-subsidisation required
3	Unit economics depend on token appreciation or external subsidy; fee volatility; business models constrained by throughput costs
4	Economics only viable via continuous external support; transaction costs prohibit adoption; no path to micropayment viability

Economic Fitness (sf) is where the entire business model of the project is evaluated. Conventional commercial analysis applies, but through the lens of whether micropayments are available as a new business model primitive — which depends entirely on whether the underlying chain can scale to accommodate micropayments. If the architecture scores high on na and ns, micropayment-enabled business models are structurally impossible, and sf collapses regardless of the business plan.

Law (H2H): Enforceability

The Law domain evaluates whether blockchain outcomes survive contact with the legal system. It examines whether the asset has standing as property, whether established remedies exist to correct disputes, and whether liability attaches to actors participating in the system.

The critical primitive is attribution. Commodity transfers in law occur through contract events represented by signatures that attribute the act of transfer to identifiable parties. A digital asset system maintains legal standing only to the extent that it preserves a continuous chain of attributable signatures linking ownership from one party to the next. Break the chain of signatures and you break the chain of title.

Score	Standing (Is)
0	Asset legally recognised as property; continuous chain of attributable signatures; parties identifiable through standard legal process
1	Standing established with standard evidentiary work; attribution chain mostly intact; identifiable through compulsory process
2	Standing arguable but requires significant evidence gathering; some attribution gaps; relies on intermediary records
3	Fragmented attribution; cross-jurisdiction complexity; title ambiguous through bridge crossings or custodial opacity
4	Non-attributable; no party to be charged; ownership record does not identify legally responsible actors; standing collapses

Score	Remedy (Ir)
0	Established legal processes available: injunction, tracing, restitution, constructive trust; property rights recoverable through courts
1	Remedies available through standard process; some operational complexity; courts have jurisdiction and precedent
2	Remedies exist but costly and slow; some dispute categories lack clear process; cross-border enforcement friction
3	Remedy practically unreliable; evidence beyond compulsion; selective enforcement only (insiders recover, others do not)
4	No meaningful legal remedy; code-as-law finality; recovery depends on discretionary insider intervention or ad hoc rollback

Score	Liability (Ip)
0	Clear liability assignment; actors identifiable and subject to jurisdiction; IP rights respected; compliance obligations understood
1	Liability established with standard legal analysis; obligations attach to identifiable parties; regulatory framework stable
2	Liability ambiguous in some areas; emerging regulatory framework; IP questions partially

	resolved; compliance costs significant
3	Liability unclear; actors may be beyond jurisdiction; money transmission questions unresolved; sanctions exposure uncertain
4	No identifiable liable party; IP violations endemic; money secrecy rules breached; sanctions non-compliance; system cannot persist legally

For a digital asset system to persist, there must be a party with standing, a court capable of granting remedy, and a party capable of bearing liability. If any element fails, legal persistence fails.

The Three Failure Cascades

Analysis of 86 blockchain implementations — 74 institutional and 12 DeFi/small-scale — reveals three distinct causal pathways to failure. Together, they account for every project failure in the dataset. Enterprise and DeFi projects trigger the same cascades through different entry points.

Cascade 1: The Architecture Death Spiral

Network.Structure → Network.Persistence → SystemState.Persistence

na → ns → sf | 75% of all failures | Entry: Network domain

This is the dominant failure mode. The project selects an architecture built on shared mutable global state, where serialisation pressure grows with write conflicts and hotspot contention. Under high-conflict workloads — ten thousand transactions interacting with the same contract or account — execution collapses toward effective sequentiality regardless of available compute. This is not a tuning problem. It is a structural ceiling imposed by state contention, independently documented in third-party reviews of enterprise blockchain implementations including the ASX CHES replacement.

The alternative — independent record-level state — allows parallelism by construction. Conflicts are localised to the specific record being spent. Transaction processing becomes a pipelined distributed system with measurable stage capacities and horizontal scaling arithmetic. These are not comparable approaches. One admits engineering. The other requires fragmentation: Layer 2s, rollups, bridges, and sharding — each of which introduces new coordination overhead, breaks atomic composability, and fragments the chain of title across jurisdictions.

Hybrid approaches attempt to bolt parallelism onto global-state architectures by requiring pre-consensus declarations of which accounts a transaction intends to touch. This introduces a coordination tax: additional broadcast messages before the actual transaction can be processed. These declarations are themselves subject to network latency, dropped packets, and race conditions. If two transactions declare non-overlapping state access but a declaration is delayed, the system must either block (adding latency) or proceed optimistically and roll back on conflict (adding complexity and failure modes). The hotspot declaration layer becomes a new vulnerability surface — all to approximate what record-level independence provides natively, without coordination overhead, by construction.

The causal chain is empirically tight: Architecture to Scalability $r=0.953$. Scalability to Economic Fitness $r=0.916$. There are zero counter-examples of a project with poor architecture achieving economic sustainability.

Case: ASX CHES Replacement. \$250M+ committed to replace the CHES clearing system with an account-based blockchain. The architecture could not support concurrent clearing volumes. After seven years, abandoned. Scores: $na=4.0$, $ns=4.0$, $sf=4.0$.

Cascade 2: Mutation Authority Collapse

SystemState.Process → SystemState.Structure → Law.Structure

sm → se → ls | 12.5% of failures | Entry: System State domain

When mutation authority is unbounded, the protocol ceases to be a fixed set of rules. Consortium governance models introduce discretion over protocol parameters, consensus rules, and execution environment changes. Too many stakeholders with competing interests produce decision-making paralysis. Execution stalls. As mutation authority expands, the legal standing of the system degrades because the system's fundamental properties — including the chain of attributable signatures — can be altered at any time by governance process.

Cascades 1 and 2 overlap almost entirely — 15 of the same 16 failed projects appear in both. This is structural, not coincidental. Unbounded mutation authority selects account-based architecture because the governance model demands shared state visibility. The mutation authority scope and the architecture choice are causally linked.

Cascade 3: The Regulatory Kill

Law.Structure / Law.Persistence | Independent of architecture

Is / Ip | 6.25% of failures | Entry: Law domain

Diem/Libra — the only project with reasonable architecture (na=2.8) that still failed. Facebook/Meta's global payment network was perceived as a direct challenge to monetary sovereignty. Central banks and regulators coordinated to kill it regardless of technical merit. SPF profile: na=2.8, ns=2.2 (adequate), but Is=4.0, Ir=4.0 (maximum legal risk).

The Regulatory Kill has a second, slower variant: account-based systems that exhibit securities characteristics face increasing regulatory classification risk under emerging frameworks. This is not a sudden kill but a slow tightening — compliance costs rise until the project becomes economically unviable, merging back into Cascade 1.

The Institutional Incompatibility Proposition

Proposition

Institutional blockchain deployments converge toward mutable governance structures, and therefore away from blockchain's defining property of protocol-constrained state transition.

Protocol immutability is architecturally achievable but institutionally avoided. The empirical pattern suggests that institutional blockchain initiatives optimise for retained discretion, not for protocol-constrained coordination.

Evidence.

Across the combined 86-project BREM dataset (74 institutional + 12 DeFi/small-scale), the mutation authority distribution is:

sm Range	Count	Interpretation
sm = 0	1 (Uniswap V2)	Protocol immutable
sm = 1	0	Narrow, formally constrained
sm = 2	8	Bounded governance
sm ≥ 3	77 (90%)	Discretionary or arbitrary

No institutional blockchain project in the enterprise dataset achieves $sm < 2$. The only $sm = 0$ project across all 86 is Uniswap V2, a permissionless DeFi protocol with deliberately immutable core contracts, no admin keys, and no upgrade mechanism. The projects at $sm = 2$ are DeFi survivors with bounded governance (Aave, Compound, MakerDAO/Sky, Solend) or institutional projects with unusually constrained architecture (mBridge). Zero enterprise projects occupy the $sm = 0$ or $sm = 1$ range.

Implication.

The result suggests that institutional blockchain is functionally a contradiction whenever institutional requirements imply retained mutation authority over core protocol rules. The governance requirements of institutional participants — regulatory compliance, operational flexibility, counterparty management, dispute resolution — demand discretion that is structurally incompatible with protocol immutability. What results is not blockchain in the architectural sense but a governed database with distributed witnesses: a system where the operator can change the rules, the data structures, the execution environment, and the network parameters without protocol constraint.

This is not a failure of implementation. It is a structural consequence of institutional incentives. Institutions optimise for discretionary control, not protocol constraint. When they adopt 'blockchain,' they systematically reintroduce the governance discretion that blockchain's defining innovation was designed to eliminate.

The Defining Boundary.

The defining boundary of BREM is not blockchain versus non-blockchain. It is constrained protocol versus retained discretion. This boundary holds across enterprise blockchain (where mutation authority selects bad architecture), DeFi (where mutation authority enables economic extraction), and hybrid models (where regulatory overlay reintroduces discretion). The variable is not branding, consensus mechanism, or token presence. It is: who can change the rules?

Blockchain is not a technology choice. It is a coordination choice. The technology merely implements the degree of constraint the participants are willing to accept. BREM measures that constraint. When the constraint is zero, the system is a database. When the constraint is total, the system is a protocol. Everything in between is measurable on the sm axis.

Empirical Evidence: Domain Severity by Status

Status	Network (M2M)	System State (H2M)	Law (H2H)
Production (n=6)	1.78	2.28	1.50
Active (n=2)	1.67	2.17	1.50
Pilot (n=1)	2.33	2.67	2.33
Discontinued (n=4)	3.45	3.57	3.63
Failed (n=16)	3.81	3.91	3.80

The Architecture–Law Nexus

The industry treats legal classification as a regulatory question. In reality, it is an architectural question.

UTXO-based architectures mirror private law’s chain-of-title structure directly: each transaction consumes previously signed outputs and creates new outputs whose ownership is defined by the next signature. The chain of digital signatures forms an evidentiary chain of title, analogous to the endorsement chain on a negotiable instrument or the recorded transfer of a commodity warehouse receipt.

Account-based architectures decouple transaction validity from attributable signatures. When projects fragment across Layer 2 networks to work around base-layer scaling constraints, bridges between chains create derivative assets with fragmented title across jurisdictions. Academic research from UCL demonstrates that the ISDA Common Domain Model cannot be deployed as a unified contract on Ethereum due to bytecode size limits, requiring modularisation that breaks the hub-and-spoke model when distributed across L2s.

The architecture-law nexus is the mechanism by which Network.Structure determines Law.Structure. Choose an architecture that breaks the chain of signatures, and you break the chain of title. Break the chain of title, and standing collapses.

Applications

The model applies wherever blockchain implementation decisions carry material risk. The underlying analytical framework is the same; what changes is the entry point and the question being asked.

Pre-Build Architecture Assessment

The highest-value application is before code is written. The Architecture Death Spiral is the dominant failure mode (75% of failures), and it is entirely predictable from the architecture choice. A pre-build assessment scores the proposed architecture against the 86-project dataset, identifies which cascade the design triggers, and surfaces regulatory classification risk through the architecture-law nexus. The question answered: does this architecture admit the scaling, economic, and legal properties the business plan requires?

Enterprise Vendor Evaluation

Enterprise blockchain decisions typically lack domain-specific evaluation criteria. Vendor claims about scalability, security, and compliance are untestable without a framework that bridges technical architecture and legal enforceability. BREM provides a structured comparison: the proposed implementation is scored against the same rubric applied to 86 existing projects across enterprise, institutional, and DeFi deployments. The evaluation is vendor-agnostic — it scores the architecture, not the brand. This applies equally to RFP processes, acquisition due diligence, and insurance underwriting.

Diagnostic Assessment

For projects already in difficulty — behind schedule, over budget, or failing to achieve adoption — the model provides root cause analysis. Most implementation pain presents as symptoms (poor performance, low adoption, regulatory friction) without diagnosis. The diagnostic question is: which cascade is the project in, and where does the constraint originate? The answer determines whether remediation is possible within the existing architecture or whether the architecture itself must change. Sunk cost pressure typically prevents this assessment from being conducted honestly without a structured external framework.

Methodology and Validation

Data

86 blockchain implementations spanning enterprise, institutional, and DeFi deployments. The enterprise dataset covers 14 sectors including financial services, trade finance, supply chain, healthcare, energy, and government. The DeFi stress test adds 12 projects including protocol collapses (Terra/Luna, FTX, Celsius), DeFi survivors (Uniswap, Aave, MakerDAO), and mid-tier protocols. Platforms include Ethereum, Hyperledger Fabric, Corda, Quorum, BSV, Hyperledger Besu, Terra, Solana, and custom implementations. Each project scored across 45 questions (9 cells × 5 questions) using domain-specific 0–4 scales.

Calibration

Expert anchor scores established for 10 representative projects across 9 anchor questions. Automated scorer uses weighted blend of expert anchors (60%) and computed baselines (40%) derived from project metadata. All scores include confidence ratings and evidence citations.

Validation

Empirical: Clean status-to-score gradient with zero overlap between production and failed distributions.

Academic: UCL dissertation on ISDA smart derivative contracts independently demonstrates architectural constraints the model predicts, including EVM bytecode limits and sequential execution constraints.

Taxonomic: ISDA Digital Asset Definitions' disruption categories map to the model's three domains, confirming the 9-cell structure captures the failure space recognised by the derivatives industry.

Appendix: Dataset Validation Summary (March 2026)

A comprehensive reclassification of all projects in the original dataset, combined with scoring of 7 additional institutional blockchain projects (LSEG DMI, BIS Project Mandala, RBA Project Acacia, NYSE Tokenized Securities, BNY Mellon Tokenized Deposits, Citi Token Services, BOJ Digital Yen Sandbox) and a 12-project DeFi stress test, produces an 86-project combined dataset. Three findings strengthen the model's empirical base.

Expanded Failure Set

March 2026 verification reveals 26 confirmed failures in the 74-project institutional dataset (35%), up from the original 16 (24%). Ten projects previously labelled as non-failures are now confirmed dead: Nigeria eNaira, Eastern Caribbean DCash, Marco Polo, Contour, eTradeConnect, Brazil DREX, TradeLens Hapag-Lloyd, Trade Information Network, BurstIQ, and Origin. The DeFi stress test adds a further 6 confirmed failures (Terra/Luna, Celsius, FTX, Iron Finance, Olympus DAO, Wonderland), bringing the combined dataset to 32 failures across 86 projects (37%). The DeFi failures alone account for over \$50 billion in destroyed value.

Predictive Accuracy

Using an overall BREM score of ≥ 2.5 as the risk threshold: 26 of 26 confirmed failures scored above this threshold. Zero false negatives. The lowest-scoring failure (Nigeria eNaira) scored 2.98. The 2.5 threshold emerges as a natural boundary: below it, projects may survive as databases with distributed witnesses; above it, the accumulated structural risk exceeds the value of distributed architecture.

The Zero-Protocol-Constraint Finding

Across the full 86-project dataset, only one project — Uniswap V2 — achieves $sm = 0$ (protocol immutable). Zero institutional blockchain projects achieve sm below 2. In the DeFi dataset, the same mutation authority patterns predict failure through a reversed cascade ($sf \rightarrow sm$ rather than $na \rightarrow ns \rightarrow sf$): economic design failures expose governance discretion, which is then exploited. 90% of the combined dataset scores $sm \geq 3$, meaning a committee or single entity can change the protocol, execution environment, data ingress rules, and network parameters without protocol-level constraint.

This is the dataset's central finding: no institutional blockchain project has achieved production deployment with protocol-constrained mutation authority. Every operational project is architecturally a database with distributed witnesses. The blockchain terminology adds cost and complexity without removing the discretion that makes blockchain architecturally distinct from a centralised system.

New Project Assessments

Project	na	sm	sf	ls	Overall	Status
LSEG DMI	3.0	4.0	3.0	1.0	2.8	Launched
BIS Mandala	2.0	3.0	3.0	0.0	1.9	PoC Phase 2

Project Acacia	3.0	3.0	2.0	1.0	1.9	Pilot Phase 2
NYSE Tokenized	3.0	4.0	3.0	1.0	2.6	Development
BNY Mellon	2.0	4.0	2.0	1.0	2.1	Operational
Citi Token Svc	2.0	4.0	2.0	1.0	2.1	Operational
BOJ Digital Yen	3.0	4.0	3.0	2.0	2.7	Sandbox

All seven new assessments confirm the pattern. sm = 4 in five of seven projects. sm = 3 in the remaining two. The two operational platforms (BNY Mellon, Citi Token Services) function as databases with distributed witnesses: processing real transactions under single-entity control on private permissioned chains. NYSE and BOJ both sit above the 2.5 risk threshold despite strong regulatory scaffolding, because their Network and System State domain scores reflect the same centralised architecture patterns seen in the failure set.

Full dataset validation details, including reclassification evidence, cascade analysis, and survivor interrogation, are provided in the companion document: BREM Dataset Validation (March 2026).

About BREM

The Blockchain Risk Evaluation Model provides structured, evidence-based risk assessment for blockchain implementations. It evaluates systems through the lens of Structure, Process, and Persistence across three coordination domains, using domain-specific scoring calibrated to protocol determinism, mutation authority, and legal enforceability.

BREM is designed to serve the full blockchain adoption lifecycle: from pre-build architecture assessment through operational monitoring to post-mortem analysis.

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